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It has been postulated that stemflow, precipitation that flows from plant crowns down along branches and stems to soils, benefits plants that generate it because it increases plant-available soil water near the base of the plant; however, little direct evidence supports this postulation. Were plants' crowns to preferentially route water to their roots, woody plants with large canopies could benefit. For example, piñon and juniper tree encroachment into sagebrush steppe ecosystems could be facilitated by intercepted precipitation routed to tree roots as stemflow, hypothetically reducing water available for shrubs and grasses. We tested whether Great Basin piñon and juniper trees use and benefit from stemflow. In a drier-than-average and wetter-than-average water year, isotopically labeled water was applied to tree stems to simulate stemflow. Both species took up stemflow, with label signals peaking and receding over 2–4 days. Despite this uptake, no alleviation of water stress was detected in the drier year. The stemflow uptake resulted in some water stress alleviation in the wetter year, specifically for piñons, which took up water from deeper in the soil profile than did junipers. Mixing model analyses suggested that stemflow was a small fraction of the water in stems (~0–2%), but an order-of-magnitude larger fraction of the stemflow was transpired in those few days after addition. These findings represent a novel demonstration of the rapid uptake and use of stemflow that infiltrates the rhizosphere, but they also prompt questions about the remaining stemflow's fate and why alleviation of water stress was so minor.

Keywords: deuterium, irrigation, isotopic enrichment, sap flow, woody plant encroachment.

Introduction

Encroachment by woody plants has been well documented in the arid and semi-arid western United States and contributes to the degradation of herbaceous and grassland habitats (Miller et al. 2008, Van Auken 2009, Wilcox and Huang 2010, Miller et al. 2019). As woody plants encroach into arid and semi-arid grasslands, there is generally a decrease in species richness and an increase in bare ground cover (Miller et al. 2000, Lett and Knapp 2005). This pattern is observed with encroachment by piñon and juniper species, where canopy cover increases result in losses of understory species (Miller et al. 2008, 2019) and an increase in bare ground. Changes in species composition can lead to changes in the water balance of these systems (Huxman et al. 2005, Wilcox et al. 2006). Moreover, as bare ground increases, so does runoff and erosion (Pierson et al. 2010), while hydraulic conductivity and soil moisture in intercanopy areas decrease (Wilcox et al. 2003, Lebron et al. 2007).

Piñon and juniper currently occupy over 18 million ha of the Intermountain West of the US and have encroached into the sagebrush steppe community, which can have significant impacts on plant communities, soil and water resources, wildlife habitat, biodiversity and fire cycles (Miller et al. 2008, 2019). It is possible that the encroachment of piñon and juniper into the sagebrush steppe community could significantly alter and reroute water in the system to the detriment of other species (Miller et al. 2008, 2019). It is well known

that taller trees can intercept light and shade lower vegetation (Breshears et al. 1997, 1998), but larger canopies also are likely to intercept more precipitation (Stringham et al. 2018). While greater interception of precipitation may result in more water losses to evaporation, thereby decreasing the amount that reaches the ground as throughfall, the greater interception may also result in greater amounts of water being routed down branches and stems as stemflow (Young et al. 1984, Martinez-Meza and Whitford 1996, Owens et al. 2006, Li et al. 2009). Hypothetically, this stemflow may disproportionately benefit the trees that generate stemflow because it reaches the soil at the tree bases where it may preferentially reach those trees' roots.

Several studies have examined the mechanism of stemflow or the fate of stemflow in the soil profile (e.g., Martinez-Meza and Whitford 1996, Li et al. 2008, 2009, Liang et al. 2011, Wang et al. 2011, Spencer and van Meerveld 2016, Allen and Conner 2021). Evidence from those studies suggests that water from stemflow may flow preferentially through soils, potentially because it is routed along roots, facilitating efficient percolation of water and increasing soil moisture deeper in the soil profile than would be expected with the infiltration of rainfall or throughfall in intercanopy spaces. Redistribution of water as stemflow can significantly increase the concentration of water near the trunk of the tree compared with bulk rainfall (Young et al. 1984, Owens et al. 2006). Owens et al. (2006) demonstrated that stemflow generated

by Ashe juniper (*Juniperus ashei*) increased soil moisture at the base of the tree by as much as 21:1 as compared with bulk rainfall soil wetting. Recent dye-tracer experiments on stemflow infiltration area at the soil surface found that stemflow infiltration is greatest near the tree trunk, but extremely localized and was $<0.23 \text{ m}^2$ per tree (Tischer et al. 2020, Llorens et al. 2022, Pinos et al. 2023), however there is still a scarcity of studies on stemflow infiltration (Van Stan and Allen 2020). Dye-tracer experiments (Spencer and van Meerveld 2016) have shown evidence of the so-called double-funneling effect of stemflow (funneling from canopy to stem, funneling from stem to macropores supporting deep percolation), but the associated destructive harvesting limits the scope of such investigations as a tree may only be excavated at a single point in time. A global review across a range of forest types found annual mean stemflow funneling ratios based on basal area to range from 3 to 37 (Levia and Germer 2015).

Even though the fraction of precipitation that becomes stemflow is generally quite small, often $\leq 2\%$ of rainfall, a global analysis across numerous forest types found stemflow to range from negligible to $>24\%$ of precipitation (Van Stan and Gordon 2018). A double-funneling effect would increase the likelihood of that water having an important fate: (i) preferential flow to enhance its availability for deep drainage and aquifer recharge (Owens et al. 2006, Liang et al. 2011, Carlyle-Moses et al. 2018), or (ii) use by vegetation because it can bypass surface soils and litter layers, making it less likely to be lost to evaporation and more likely to benefit plants (Young et al. 1984). Many studies suggest or argue that stemflow water is important for preferentially routing water to roots, potentially providing key moisture to supporting physiological activity in trees and shrubs when upper soil layers are dry (Martinez-Meza and Whitford 1996, Owens et al. 2006, Li et al. 2008, 2009, Wang et al. 2011); however, no studies have successfully tracked the fate of stemflow water from the soil through the plant. Thus, hypotheses regarding stemflow water traveling through soils along root channels or quickly becoming available for uptake by the plants that generate it or stemflow (Martinez-Meza and Whitford 1996, Liang et al. 2011) remain undertested.

Stable isotope ratios of hydrogen (δD) and oxygen ($\delta^{18}\text{O}$) in water are ideal tracers for tracking water movements from precipitation into soils and from soils into plants, if precipitation isotopic compositions differ substantially from those of other potential soil water or groundwater sources (Dawson and Ehleringer 1991, Flanagan and Ehleringer 1991, Flanagan et al. 1992). Although it had long been assumed that water extracted from a plant typically bears an isotopic composition that reflects the mixture of isotopic compositions of water sources used by that plant (i.e., there is no fractionation upon uptake or fractionation), a growing body of evidence has shown that isotopic offsets can occur during root uptake and/or extraction (Lin and Sternberg 1993, Ellsworth and Williams 2007, Zhao et al. 2016, Chen et al. 2020, Barbata et al. 2022); however, those offsets (i.e., systematic bias) will have minor consequences if the offsets are small relative to the signal examined; that is to say, they are less important when potential sources have highly distinct isotope ratios (Allen and Kirchner 2022). For tracing differences in the percolation or root uptake of stemflow versus rainfall or throughfall, naturally occurring differences between stemflow and throughfall isotopic compositions are detectable but they rarely exceed 1.0‰ $\delta^{18}\text{O}$ (Allen et al. 2017); these small separations among

natural source signatures can complicate the interpretation of stemflow contributions to transpiration (Snelgrove et al. 2020). To circumvent issues that arise from uptake and extraction artifacts resulting in high noise-to-signal ratios, applying isotopically labeled water is an appealing option. The fate of experimentally applied water that is enriched in heavy isotopes can be tracked because it is isotopically distinct from naturally occurring environmental waters (von Freyberg et al. 2020, Snyder et al. 2022). Notably, isotopically enriched water can be used to experimentally determine if plants respond to watering events (Williams and Ehleringer 2000, Schwinning et al. 2002, Snyder and Williams 2003, 2007, West et al. 2007a, Kulmatiski et al. 2017, Aguzzoni et al. 2022). In this study, isotopically enriched stemflow was applied to trees to determine if xylem water becomes isotopically enriched as a result of using the stemflow water.

Our objective was to determine if Singleleaf piñon (*Pinus monophylla* Torr. & Frém.) and Utah juniper (*Juniperus osteosperma* (Torr.) Little) used stemflow water. We directly tested the benefits of stemflow to trees transporting the stemflow, however we did not test the use of stemflow by other vegetation. While stemflow could have a larger ecosystem role, a first step toward identifying such a role is to determine whether stemflow is a substantive component of the water used by piñons and junipers and whether it improves water status of the plant that generated the stemflow. In addition, understory vegetation is sparse or nonexistent under the crowns of these trees and in the intercanopy area, which is expected in this highly encroached piñon and juniper woodland where trees become dominant (Miller et al. 2008, 2019). We hypothesized that these trees would use stemflow water, and that the use of stemflow water would result in a physiological benefit to the plant, specifically as an increase in sap flow or increase in plant water potential. We hypothesized that this effect would be more substantial for upland trees than for valley trees due to water being more limited on the hillslope. We also hypothesized that piñon would be more likely to use stemflow water, due to the fact that it is less drought tolerant than juniper (West et al. 2007b) and thus may be more opportunistic in taking up water when it is available. We hypothesized that these trees would use stemflow water, and that the use of stemflow water would result in a physiological benefit to the plant, specifically as an increase in sap flow or decrease in plant water potential. We hypothesized that this effect would be more substantial for upland trees than for valley trees due to water being more limited on the hillslope. We also hypothesized that piñon would be more likely to use stemflow water than juniper, due to the fact that it is less drought tolerant than juniper. This study builds on a previous study that was conducted at the same sites, which found that the presence of piñon and juniper reduced net precipitation by an average of 44.6% ($\pm 27.0\%$), depending on the simulated storm size and intensity, which ranged from 2.2 to 25.6 mm h^{-1} (Stringham et al. 2018). That study also established that stemflow increased with larger storms, ranging from 0 to 1.9 mm , summing to $\sim 2\text{--}4\%$ of the applied precipitation (Stringham et al. 2018). While throughfall ($\sim 58\%$ of precipitation) influxes far exceed those of stemflow here, and thus they must be the primary source recharging the soil-water storages that supply plants, this study tests the role of stemflow to trees in isolation of throughfall, to test whether stemflow is even notably important in those conditions.

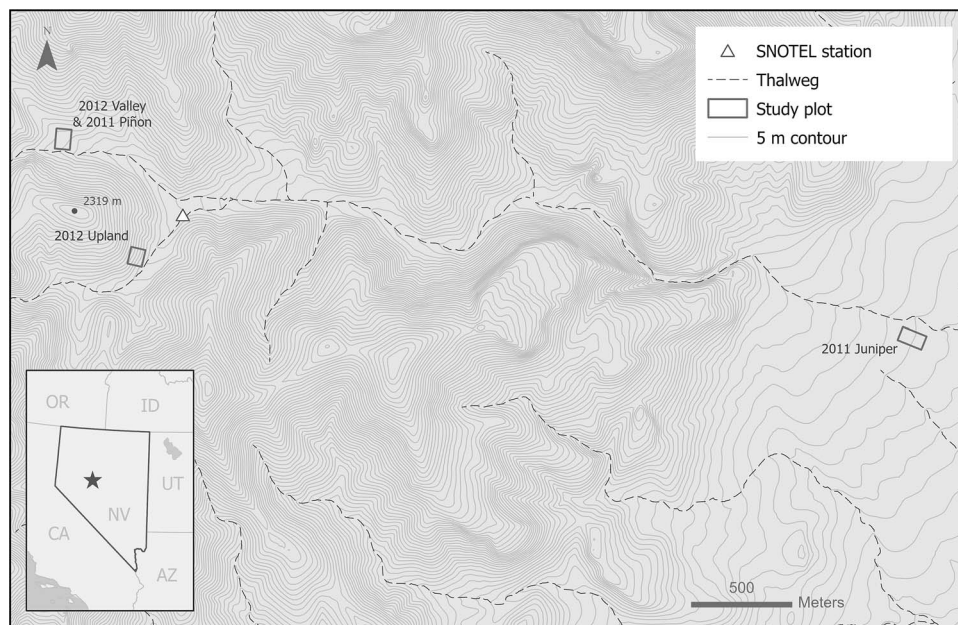


Figure 1. A map showing the locations of the sampling sites in 2011 and 2012 and the SNOTEL station with elevation and watershed characteristics.

Materials and methods

This study was conducted in Porter Canyon Experimental Watershed (PCEW), which is located in Lander County, Nevada on the eastern side of the Desatoya Range in the Great Basin ($39^{\circ} 28' N$; $117^{\circ} 37' W$, 2195 m above sea level). PCEW lies within a semi-arid mountain range dominated by snowfall and is classified as a cold desert (Weisberg et al. 2007). The study area is on private land and surrounded by land owned by the Bureau of Land Management (Dept of the Interior). Long-term mean annual precipitation based on 30 years of data (1991–2021) was 303 ± 6.7 mm per year, obtained from PRISM (PRISM Climate Group, Parameter-elevation Regression on Independent Slopes Model, <https://prism.oregonstate.edu>). PRISM precipitation totals for water year 2011 were 425 mm, and 218 mm for water year 2012 (Daly et al. 1994; PRISM Climate Group, <https://prism.oregonstate.edu>). PCEW is within the rain–snow transition zone. As global temperatures have increased, more precipitation has been observed as rainfall instead of snow in the Great Basin, with a decline in the snow fraction of 1.9% per decade during the period 1916–2003 (Safeeq et al. 2016). An NRCS soil survey determined that the soils in the canyon are derived from the weathering of tuff and other volcanics with regolith present at 50–80 cm. Soil textures are extremely cobbly fine sandy loam (35% of the area), extremely cobbly loam (25%) and very gravelly loam (25%). The soil at the mouth of the canyon is alluvium of an alluvial fan remnant derived from a mixed source of rock with a duripan layer present at 35–50 cm, and the next restrictive feature at over 200 cm deep. Soil textures are gravelly loam (55%) and gravelly fine sandy loam (30%) (Soil Survey Staff Natural Resources Conservation Service USDA, n.d.).

The main plant communities within the study area of Porter Canyon include piñon-juniper, sagebrush steppe and wet meadow areas where groundwater springs are present. The hillslopes are now dominated by piñon and juniper, while the understory is primarily composed of Sandberg's bluegrass (*Poa secunda* J. Presl). Mountain big sagebrush (*Artemisia tridentata* Nutt. subsp. *vaseyana* (Rydb.) Beetle) is found lower

on the hillslopes (Lossing 2012), and throughout the valley floor of the canyon, along with two species of rabbitbrush (*Chrysothamnus viscidiflorus* (Hook.) Nutt. and *Ericameria nauseosa* (Pall. ex Pursh) G.L.Nesom & G.I.Baird).

In 2011, a piñon site and a juniper site were selected. The piñon site was located on the valley floor of the canyon in Porter Canyon. This area is a relatively flat, open area where trees have encroached into a sagebrush area. The juniper site was located outside Porter Canyon near the canyon mouth. This area was also relatively flat and open, and Porter Creek intermittently flows nearby. In 2012, two sites within Porter Canyon were selected for this study: a valley site (which was the same site as the piñon site in 2011) and an upland hillslope site, both sites contained piñon and juniper. The upland site was located on a hillslope with a slope of $\sim 20^{\circ}$. In sum, three sites were used over the 2 years of the experiments and different trees were used in 2011 and 2012.

Environmental measurements

We used data from the SNOTEL station located within PCEW. Relative humidity and temperature were measured at ~ 3 m height using a Vaisala HMP45C sensor (Vaisala, Woburn, MA, USA). Precipitation accumulation was recorded using a pressure transducer.

Stemflow experiment

An experiment was conducted to determine the amount of stemflow used by these species. July and August were chosen to maximize the chance of detecting the labeled water in the trees by having relatively dry soils. Six piñons were selected at the valley site in 2011, and six junipers were selected near the mouth of the canyon (Figure 1). Seven piñons and six junipers were selected at each the valley and upland sites in 2012 (Figure 1). The piñons selected in 2011 were of two size classes; we hereafter refer to small (7–16 cm stump diameter) piñons, of which there were two, and medium piñons (17–32 cm stump diameter), of which there were four. Stump diameter was measured at ~ 50 cm above ground. Junipers

Table 1. Mean $\pm$ 1 SE stump diameter, estimated mean $\pm$ 1 SE sapwood area of trees taken 50 cm from the ground surface and mean $\pm$ 1 SE of tree rings for species size classes at each site. JUOS, *Juniperus osteosperma*; PIMO, *Pinus monophylla*.

Site	Species	Size class	Mean stump diameter at 50 cm (cm)	Estimated mean sapwood area (cm ²)	Mean annual rings
Upland	JUOS	Med	20.15 $\pm$ 2.75	399.22 $\pm$ 205.10	114.50 $\pm$ 14.50
Upland	JUOS	Sm	9.93 $\pm$ 1.29	72.6 $\pm$ 16.74	76.25 $\pm$ 14.20
Upland	PIMO	Med	25.10 $\pm$ 1.24	222.70 $\pm$ 19.92	96.25 $\pm$ 6.61
Upland	PIMO	Sm	10.63 $\pm$ 1.10	52.09 $\pm$ 8.64	68.33 $\pm$ 6.01
Valley	JUOS	Med	24.23 $\pm$ 0.26	442.18 $\pm$ 61.24	85.67 $\pm$ 12.17
Valley	JUOS	Sm	16.50 $\pm$ 1.30	189.28 $\pm$ 26.85	78.50 $\pm$ 4.17
Valley	PIMO	Med	24.35 $\pm$ 2.50	284.10 $\pm$ 31.16	63.67 $\pm$ 10.04
Valley	PIMO	Sm	9.83 $\pm$ 0.52	74.28 $\pm$ 15.43	41.33 $\pm$ 8.01

in 2011 did not fall into these categories and their diameters ranged from 23 to 44 cm. In 2012, there were three small and four medium piñons at each site, three small and three medium junipers at the valley site and four small and two medium junipers at the upland site. Small trees at the upland site had average diameters of 9.9 ± 1.3 cm for juniper and 10.6 ± 1.1 cm for piñon, while the average for medium trees was 20.2 ± 2.8 cm for juniper and 25.1 ± 1.2 cm for piñon. At the valley site, small trees were 16.5 ± 1.3 cm for juniper and 9.8 ± 0.5 cm for piñon, and medium trees were 24.2 ± 0.3 cm for juniper and 24.4 ± 2.5 cm for piñon (Table 1). Other than a few large old growth trees that were not studied, these size classes represented the natural variation found in piñon and juniper stands at this study site.

Water isotopically labeled with deuterium (δD) was applied to each tree to simulate stemflow expected to result from a 20 mm rainfall event. In 2011, the amount of stemflow applied to the trees was determined by measuring the amount of stemflow generated by each specific tree during the simulation of a 20 mm rainfall event (Stringham et al. 2018). In 2012, stemflow volume was calculated using a more generalized approach, using the following predictive model based on data collected from rainfall simulations done in summer 2010 and 2011 at PCEW (Lossing 2012):

$$SF = 0.04(0.421 + 0.875S - 0.021A_C - 0.020S^2 + 1.68e^{-6}SA_C + C)^2 \quad (1)$$

where SF is the quantity of stemflow to be added in liters, S is the storm size in mm (20 mm for this experiment), A_C is the canopy area in m² and C is a coefficient for crown shape. For cone shaped canopies C is -0.97 , and for expanded paraboloid (round-topped cylinder) shaped canopies C is 0.97 . This equation is a modification of the Best Fit Model with $R^2 = 0.795$ (Lossing 2012), where stemflow was determined to be 4% of total redistribution (throughfall plus stemflow) on average, for storm sizes ranging from 19 to 25 mm (20 mm storm class). This estimation is in line with previous research on stemflow (Young et al. 1984, Owens et al. 2006, Li et al. 2009). The applied stemflow amounts ranged from 1.1 to 17 L in 2011 and from 2.7 to 4.1 L in 2012, varying with tree size. While this was the amount modeled to be associated with 20 mm storms, natural inter-storm variations are substantial so similar amounts of stemflow could occur with much larger or smaller size storms.

Deuterium was added to well water obtained at the site. Large buckets were used to mix the water and samples of the

mixed water were taken to confirm isotopic enrichment. In 2011, the labeled water to be applied as stemflow was mixed to have an enrichment of $+200\text{‰}$ δD . In 2012, the labeled water was mixed to be $+400\text{‰}$ δD . The isotopically labeled water was applied using standard 1.27 cm garden soaker hose wrapped around the base of the trunk to simulate stemflow. The soaker hose was attached via plastic tubing to a plastic bucket that was suspended from a branch. Labeled water was added to each bucket and water flowed down the tree trunk and infiltrated the soil near the bole to simulate stemflow. The stemflow-addition process took approximately between 30 and 60 min. The methodological approach is not sensitive to the use of a stronger label (i.e., in the second year); however, using a stronger label increases the signal-to-noise ratio and thus increases the likelihood of stemflow being detected in the stem xylem.

Pre-dawn water potentials were measured, and plant stem samples were collected on the day before stemflow was applied to establish 'Day 0' data for the experiment. That evening (Day 1) and the following evening (Day 2), plant stem samples were collected for stable isotopic analyses of xylem water. Stem samples were also collected for the next three mornings (Day 2 through Day 4). Soil samples were collected on Day 1 from open spaces within the plot at increasing depths. Predawn xylem water potentials were measured every night through Day 4 to quantify changes in water availability as experienced in the rhizosphere. Midday xylem water potentials were measured every afternoon for Day 1 through Day 3 (see specific methods below).

In 2011 the stemflow experiment was conducted from 17–21 July at the piñon site and from 31 July to 4 August at the juniper site. In 2012 the experiment was conducted from 9 July to 13 July at the upland site and from 30 July to 3 August at the valley site.

Isotope sampling

Plant stem samples were collected from sunlit branches at various locations around each tree for analysis of stable isotope composition of xylem water. Selected branch sections were suberized and located at least 10 cm more proximal to the stem than the nearest foliage on that branch. Bark was not removed but we assume this to have minimal impact on the signals interpreted given recent developments in our understanding of phloem–xylem exchange and its effects on stem stable isotope ratios (Nehemy et al. 2022). Soil samples were collected with a soil auger. In 2011 soils were sampled at 0–5, 5–10, 20–25 and 45–50 cm. In 2012, soil samples were collected at 5–10, 10–15 and 20–25 cm. A portion of each soil

sample was collected to be analyzed for the stable isotopic composition of soil water. The remaining soil was sealed in a soil tin for gravimetric water content analysis. All samples for isotope analysis were collected in borosilicate glass vials and sealed with parafilm. Soil and plant samples were stored in a dry cooler away from sunlight until transport to the lab where they were placed in a freezer until isotope extractions were performed.

Water was extracted from plant and soil samples by cryogenic vacuum distillation (Ehleringer and Osmond 1989). Water from plants, soil and labeled stemflow were analyzed for oxygen ($\delta^{18}\text{O}$) and hydrogen (δD) stable isotopes using a Los Gatos LWIA Wavelength-Scanned Cavity Ring-Down Spectrometer at the University of Wyoming Stable Isotope Facility. While analysis of water from plant samples by laser spectroscopy may introduce errors (see Schultz et al. 2011 and West et al. 2011), both pre- and post-stemflow addition samples would be affected and thus these represent uncertainties rather than directional biases; our analytical methods are sensitive to those uncertainties and thus associated effects of spectral contamination are implicitly accounted for in our analyses and inferences.

Plant water status

Predawn (Ψ_{pd}) and midday shoot water (Ψ_{md}) potentials were measured using a Scholander-type pressure chamber (PMS, Corvallis, OR, USA). Ψ_{pd} was measured between 1:30 and 3:00 a.m. Ψ_{md} was measured between 1:00 and 2:00 p.m. Distal twigs were cut from a random position on the tree between chest and eye level using sharp clippers, and samples were sealed in whirl-pack bags. Water potential measurements were made within 2 min of cutting the twigs. For Ψ_{md} measurements, stems were kept in a shaded area until measured to minimize water loss.

Sap flux measurements

Granier-style thermal dissipation probes were used to measure sap flux on all stemflow trees during the stemflow experiment. The one exception was 2011 juniper: all six trees were instrumented, but due to technical difficulties there was no Day 0 data, and two of the six trees were excluded from this analysis, due to problems with probe installation.

In 2011 at the upland site, piñons were instrumented with two sets of 2 cm Granier-style sap flow probes. Junipers were instrumented with two sets of 1 cm Granier-style sap flow probes. In 2012 all trees were instrumented with 2 cm probes, as tree core measurements from late 2011 indicated that sapwood depth was greater than 2 cm for all trees. The probes were connected to a CR1000 datalogger where measurements were logged every 15 min with 1-min sampling intervals. Two sets of probes were installed radially into the xylem at approximately north-south orientation, depending on branch architecture and spaced 15 cm axially with the top probe ~50 cm off the ground. Each probe contained a copper-constantan thermocouple at half the length of the probe. The upstream (lower) probe was an unheated reference probe, and the downstream (upper) probe was wrapped with a constantan heater coil that received constant power. The two probes were wired together, and a temperature difference was calculated between the two. The temperature difference between the probes is related to sap flow density by

the following empirical equation (Granier 1985, 1987):

$$J_s = 0.0119 \left(\frac{\Delta T_0}{\Delta T} - 1 \right)^{1.23} \quad (2)$$

where J_s is sap flux density ($\text{g cm}^{-2} \text{s}^{-1}$), ΔT is the temperature difference between the heated and unheated probes, and ΔT_0 is the temperature difference obtained under zero-flow conditions (West et al. 2008). Sap flow measurements were analyzed to determine ΔT_0 using a script in the R program statistical package, that found the maximum ΔT_0 values each day from 10 p.m. to 10 a.m. Data were recorded as 15-min averages based on 1-min measurements. Data from the north and south aspects were averaged to account for azimuthal variation. This was then scaled up to a 15-min time frame and 15-min intervals were summed to calculate hourly data. Hourly data were summed over each 24-h period to calculate daily data. Hourly sap flux density was multiplied by sapwood area to scale up to the whole tree and hourly values were summed to get daily sap flow volume estimates of the water used by each tree. Sapwood area was determined by coring trees to determine the depth of sapwood and measuring the diameter of the trees at 50 cm above the ground and creating an allometric relationship to predict sapwood length (Table S1 available as Supplementary data at *Tree Physiology* Online).

Data analysis

A mixed-model repeated measures analysis of variance (ANOVAR) was used to determine if trees had significant changes in δD values after irrigation that would indicate use of stemflow irrigation. At each site, ANOVAR was used to determine the effects of species, size class and sampling date on δD values. When significant interactions were found, differences between δD values were determined for each sampling date by testing the one-degree of freedom hypothesis (i.e., slice tests) of difference between species or size class. The 2011 piñons were compared by size using ANOVAR. The 2011 junipers were analyzed using mixed-model ANOVAR with only sampling date since there was no species or size class comparison.

The initial δD values (Day 0) for each species were used to determine what water sources trees were using before the experiment. The values were averaged by size class if size class was found to be significant in the above analyses, and otherwise they were averaged by species. These mean values were analyzed using IsoSource software (Phillips and Gregg 2003) to determine the range of feasible proportions of water used from different soil depths.

A two-compartment linear mixing model was used to calculate the proportion of stemflow irrigation used by the tree compared to bulk soil water. The model was derived from (Snyder and Williams 2007):

$$f = \frac{\delta\text{D}_{\text{plant}} - \delta\text{D}_{\text{day0}}}{\delta\text{D}_{\text{pulse}} - \delta\text{D}_{\text{day0}}} \quad (3)$$

where f is the fraction of water derived from the labeled pulse of stemflow water, $\delta\text{D}_{\text{plant}}$ is the value of the xylem water of each tree from the day with the largest change in isotopic signature, $\delta\text{D}_{\text{day0}}$ is the label of the tree on Day 0 of the experiment, and $\delta\text{D}_{\text{pulse}}$ is the value of the labeled stemflow

water that was applied to each tree. If negative mixing values were obtained, they were left in the analysis. The effect of size class was tested with a comparison of means. The values for f were analyzed using a one-sided t -test to determine if the values were significantly different from zero, where zero indicates no stemflow use.

We also used the stable isotope data to estimate the fraction of added stemflow that becomes stem water (to be transpired). This is a forward analysis, addressing the question of how the stemflow will be split between uptake and other fates, which differs from mixing model analyses such as those described in the previous paragraph, which are backward analysis asking where stem water came from; relationships between mixing analyses and end-member splitting are discussed extensively elsewhere (Kirchner and Allen 2020). A primary difference is that this splitting analysis involves water-balance closure in addition to isotope mass-balance closure. We multiplied the single-day-maximum fraction of stemflow in xylem (i.e., the mixing-model solution) by the sap flow on that day (in L per day), to estimate the volume of stemflow transpired on that day. The volume of stemflow transpired on that day was divided by the volume of labeled stemflow added to each tree to quantify the fraction of the added stemflow that went toward supplying the trees. The resulting values were analyzed using one-sided t -tests to determine if the values were significantly different from zero, and two-sided t -tests to determine whether the values differed by species or hillslope position.

Water potentials were analyzed using a one-sided t -test to determine if values were significantly different from Day 0 (Ψ_{pd}) or Day 1 (Ψ_{md}) to indicate an alleviation of plant water stress. Total daily volume of water transpired was analyzed for each site, species and size class; size class was included due to the inherent differences when scaling to the whole tree. A repeated measures ANOVA was used with daily volume as the response variable, day of the experiment (i.e., time) as the within factor variable, and tree identification number as the repeated factor.

Results

Environmental data

The mean relative humidity in 2011 was 27.3% and 25.5% in 2012. Peaks in relative humidity were 92.8% in 2011, and 98.0% in 2012 on days that it rained. The mean air temperature in July 2011 was 20.1 and 20.5 °C in July 2012, with a maximum of 31.6 °C in both years. Data from PRISM showed that cumulative precipitation at the end of July was above the average of 300 mm in water year 2011 (372 mm), and below average in water year 2012 (191 mm). There were no significant rain events at the sampling site in July 2011. The events that occurred from 1 August 2011 to 3 August 2011 were largely isolated to the area around the weather station and did not affect soil moisture content at the juniper site (Figure 2a). Vapor pressure deficit (VPD) calculated from RH and temperature sensors within the canopy did show that VPD decreased during those rainfall events (Figure 2a). In 2012, a rainfall event occurred on the last day of the upland experiment, but after sampling was completed. However, VPD was lower on the last day of sampling (Figure 2b). The valley experiment was conducted during a rainless period (Figure 2b).

Soil moisture (Table 2) was highest at the piñon site in 2011, with the greatest value (12.4%) seen at 10–15 cm, with values decreasing with depth. The 2011 juniper site had much lower values of soil moisture and was driest at 10–15 cm (4.5%) and increased with depth. In 2012 the valley site had the greatest water content, again with the greatest moisture content at 10–15 cm (8.2%) and decreasing at 20–25 cm (6.5%). The upland site had low water content at the surface (4.3%) that increased with depth to 9.5% at 20–25 cm.

Stable isotopes

The soil water in 2011 showed a more negative δD value with depth, but little change below 25 cm. Soil water δD at the piñon site were -116.0‰ in the 5–10 cm sample, -132.1‰ at 20–25 cm and 134.3‰ at 45–50 cm. At the juniper site the soil water δD of upper sample was -96.4‰ , and -133.7‰ at 20–25 cm. The 45–50 cm sample was -142.5‰ (Figure 3a). IsoSource found that junipers used a mix of water from all soil depths with a wide range of feasible solutions (Table 3), whereas IsoSource yielded more definitive results for piñon, indicating that they obtained between 95% and 100% from 50 cm soil depth.

The soil water at the 2012 sites also showed a decrease in δD values from 5 to 25 cm. The upper soil water sample was -104.8‰ at the upland site, and -125.2‰ at 20–25 cm. The juniper xylem water at the upland site on Day 0 had a mean δD value of -123.2‰ and piñon xylem water was -120.0‰ , both at a value corresponding to a soil depth between 15 and 25 cm. At the valley site the soil water showed an even greater decrease in δD values with depth, ranging from -45.7‰ at 5–10 cm to -109.5‰ at 20–25 cm (Figure 3b). Tree xylem water at the valley site also had δD values that corresponded to 15–25 cm soil depth on Day 0 with the mean for juniper at -94.8‰ and piñon at -106.1‰ (Figure 3b). IsoSource at the upland site found the piñon and juniper were likely using water with signatures between those of the shallower and deeper soil depths (which only went down to 25 cm in 2012; these depths reflect the deepest that could be excavated with tools at the time, and thus readers should note that these values do not capture the full depth-distribution of relative water uptake. At the valley site, medium piñon was using a source more negative than any of the sampled sources, while small piñon and small and medium juniper had a wide range of feasible solutions from the three soil depths.

Stemflow water use and the time-series of δD

In 2011, there was no significant effect of size or sampling date on piñon xylem water δD and size class was discarded. The xylem water of the species were significantly different with juniper being more positive than piñon ($P < 0.0001$). No isotopic enrichment from the labeled stemflow was seen for piñon on Day 1, but there was a slight isotopic enrichment on Day 2 to -132.8‰ and -130.4‰ for piñon xylem water, respectively. For juniper xylem water there was a trend for sampling date to be significant at $P = 0.084$. Juniper xylem water exhibited a more positive δD value on Day 1 of -119.8‰ , which declined only slightly on Days 2 and 3 (Figure 4).

In 2012 at the upland site, there was a significant interaction of species and sampling date ($P = 0.01$) on xylem water δD values; there was no effect of size class ($P = 0.48$), therefore size was discarded from the analysis. The initial δD value for juniper xylem water on Day 0 was -123.2‰ , and the δD value

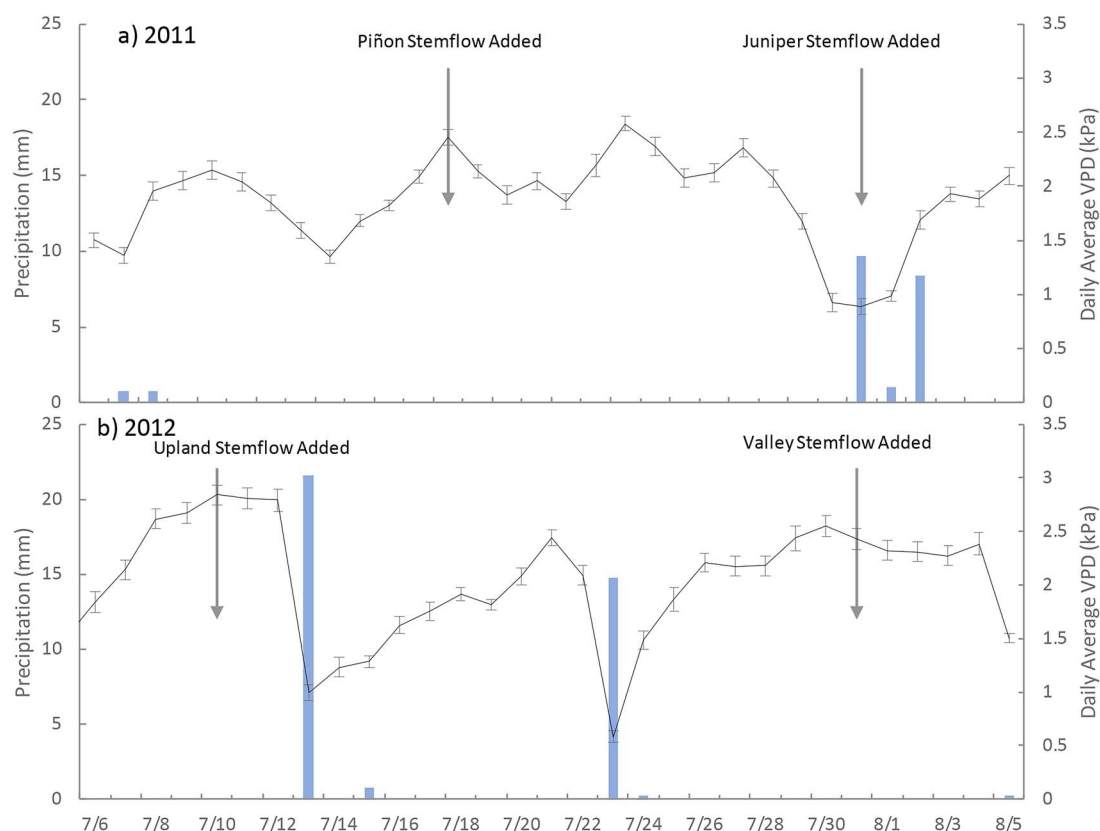


Figure 2. Daily precipitation (bars) recorded by the Natural Resources Conservation Service weather station over the duration of the experiments for 2011 (a) and 2012 (b). Daily mean ± 1 SE VPD calculated from sensors within the tree canopy is shown as the line. Arrows indicate days that stemflow was added.

Table 2. Mean ± 1 SE gravimetric soil water content.

Depth (cm)	2011 Piñon θ_g (%)	2011 Juniper θ_g (%)	2012 Upland θ_g (%)	2012 Valley θ_g (%)
5–10	9.4 ± 2.2	5.2 ± 0.5	4.3 ± 0.7	8.0 ± 1.4
10–15	12.4 ± 1.7	4.5 ± 1.0	6.7 ± 0.2	8.2 ± 1.0
20–25	10.9 ± 1.4	6.0 ± 0.5	9.5 ± 0.4	6.5 ± 0.2
45–50	8.0 ± 0.7	6.1 ± 0.8	—	—

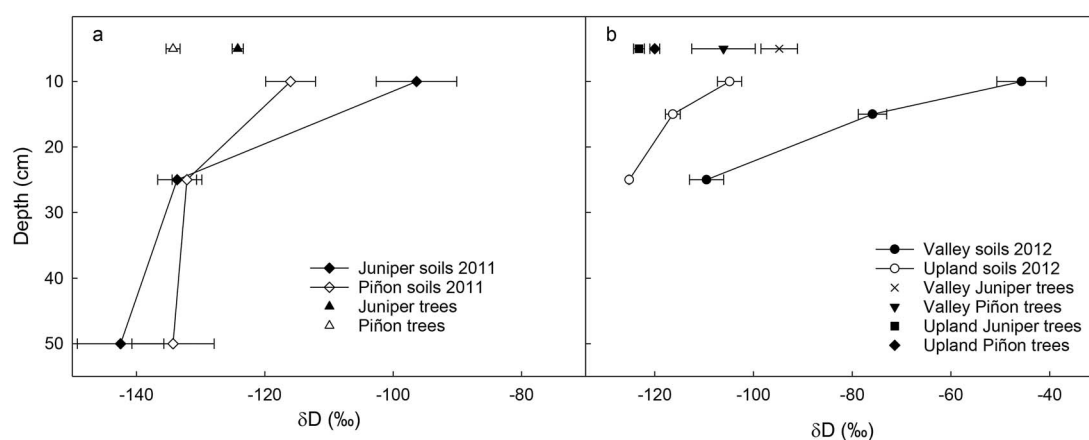


Figure 3. Mean ± 1 SE deuterium composition of soil water and tree xylem water on Day 0 in 2011 at the piñon site and juniper site (a) and in 2012 at the valley site and upland site (b).

Table 3. The IsoSource range of feasible proportions of water from each soil layer for each species and site, using the classifications for size class in the valley, where ANOVA revealed a significant difference between tree size within a species. N/A is when the tree δD value was more negative than any source, thus the model was not able to find any feasible solutions. JUOS, *Juniperus osteosperma*; PIMO, *Pinus monophylla*.

Year	Site	Species (size class)	<i>n</i>	10 cm	25 cm	50 cm
2011	Piñon	PIMO	6	0	0–0.05	0.95–1.00
2011	Juniper	JUOS	6	0.25–0.39	0.02–0.75	0–0.59
				10 cm	15 cm	25 cm
2012	Valley	PIMO (sm)	3	0–0.28	0.01–0.54	0.46–0.71
2012	Valley	PIMO (med)	4	N/A	N/A	N/A
2012	Valley	JUOS (sm)	3	0–0.19	0.17–0.37	0.63–0.80
2012	Valley	JUOS (med)	3	0–0.27	0.01–0.05	0.48–0.72
2012	Upland	PIMO	7	0–0.26	0–0.59	0–0.41
2012	Upland	JUOS	6	0–0.08	0–0.20	0.8–0.92

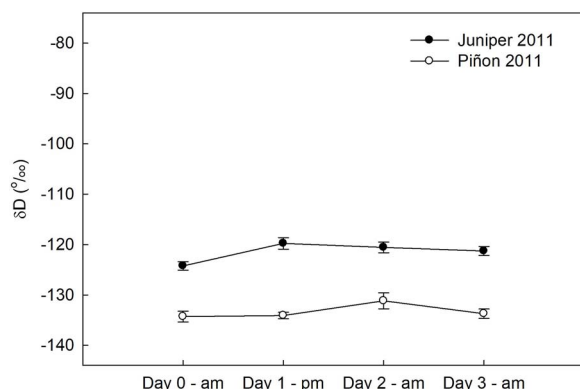


Figure 4. Mean ± 1 SE tree xylem water deuterium composition before (Day 0) and after (Days 1–3) the application of stemflow measured in 2011.

for piñon xylem water on Day 0 was -120.0‰ . The δD values of piñon and juniper xylem water started to diverge on Day 1, with a significant difference seen on Days 2 and 3 ($P = 0.04$, Tukey's post-hoc pairwise comparison). The greatest isotopic enrichment resulting from labeled stemflow water for juniper xylem water was on Day 2 with a δD value of -119.6‰ . The greatest isotopic enrichment for piñon xylem water was on Day 3 with a δD value of -114.6‰ (Figure 5a).

In 2012 at the valley site there was a significant interactive effect of species and sampling date ($P = 0.04$) and a significant interaction of species and size ($P = 0.02$). In general, small trees and medium juniper xylem water had a more positive δD value than medium piñon xylem water (Figure 5b). Small juniper xylem water values were -97.6‰ on Day 0 and small piñon xylem water values were -91.4‰ on Day 0. Medium juniper xylem water values were -92.0‰ on Day 0 and showed isotopic enrichment through Days 1 and 2 to approximately -84.5‰ from uptake of labeled stemflow water. Medium piñon xylem water values were -114.1‰ on Day 0 and exhibited a slight isotopic enrichment through Day 2 to about -111.4‰ (Figure 5b). There was a significant difference in the δD values of medium piñon and medium juniper xylem water in the valley ($P = 0.02$), with the medium juniper xylem water δD values being more positive.

Isotope mixing model

A linear mixing model (Eq. (3)) determined the proportion of xylem water composed of stemflow. An ANOVA model

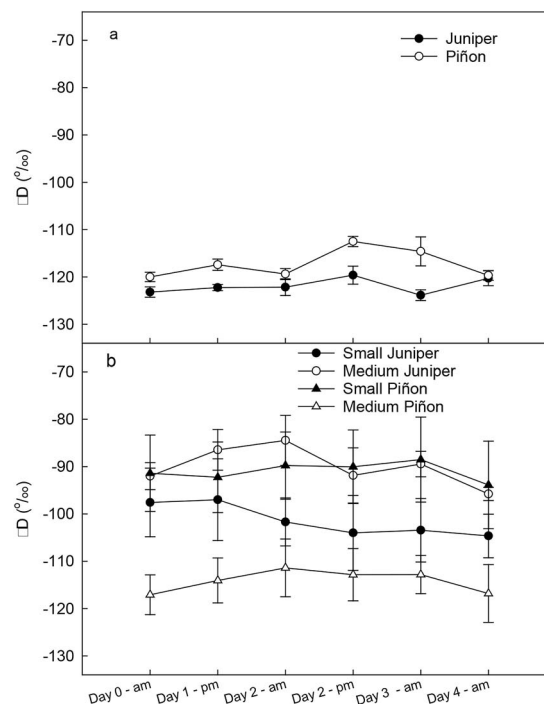


Figure 5. Mean ± 1 SE tree xylem water deuterium composition before (Day 0) and after (Days 1–4) the application of stemflow for upland trees (a) and valley juniper and valley piñon (b) measured in 2012.

determined there was no significant effect of size class on that proportion for either species, therefore size was discarded in the statistical test. One-sided *t*-test were used to determine if the fraction of stemflow water was significantly different than zero ($P_{\text{crit}} < 0.05$, Table 4). In 2011, the fractional contribution to xylem water of stemflow water was $1.00 \pm 0.23\%$ for piñon, and $1.68 \pm 0.38\%$ for juniper. In 2012 at the valley site, the fractional contribution of stemflow water was $1.03 \pm 0.36\%$ for piñon; in contrast the fractional contribution of stemflow water for juniper was not significantly different than zero. For the upland site in 2012 the fractional contribution of stemflow water for piñon and juniper was $1.78 \pm 0.3\%$ and $1.01 \pm 0.11\%$, respectively.

Sap flow

Total daily sap flow volume showed no significant differences with time (i.e., day of experiment) for any site, species and size

Table 4. The percent of water derived from labeled stemflow irrigation (f) for each species and site, using the classifications for size class in the valley, where ANOVA revealed a significance between tree size within a species. P -values that are noted with (*) indicate results that are significantly different from zero at $\alpha = 0.05$.

Year	Site	Species (size class)	n	f (%)	P -value
2011	Piñon	PIMO	6	1.00 ± 0.23	0.004*
2011	Juniper	JUOS	6	1.68 ± 0.38	0.003*
2012	Valley	PIMO	7	1.03 ± 0.36	0.015*
2012	Valley	JUOS	6	0.48 ± 0.73	0.27
2012	Upland	PIMO	7	1.78 ± 0.43	0.003*
2012	Upland	JUOS	6	1.02 ± 0.11	0.0001*

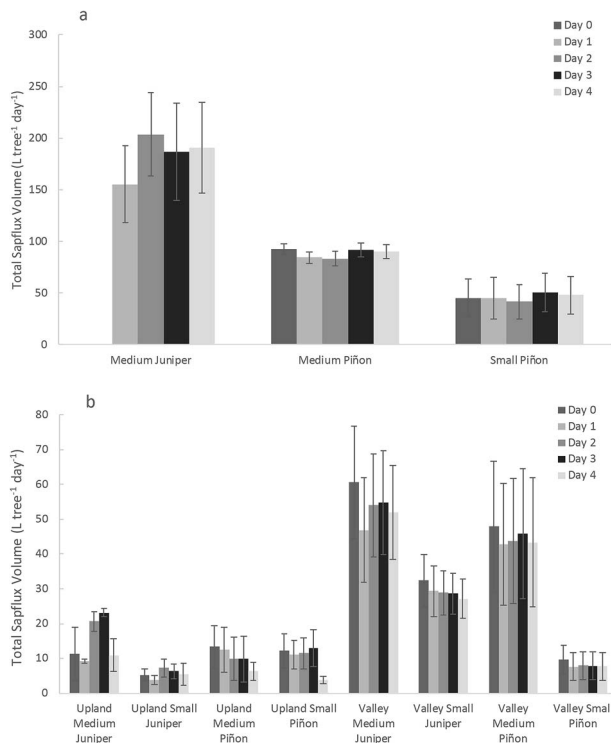


Figure 6. Mean $\pm$ 1 SE daily sap flow volume observed during the stemflow experiment in 2011 (a) and 2012 (b).

combination, despite uptake of stemflow water (Figure 6). For all site, species and size combinations the effect of time was not significantly different ($P \geq 0.09$).

Fraction of label used

While the fractions of water in xylem sourced by the added stemflow were seemingly small, when the isotope data were reframed to infer the fraction of added stemflow that becomes transpired, we saw that a substantial fraction of the added stemflow is transpired (Figure 7). In one tree, the data suggest that transpiration may use as much as 79% of the added stemflow pulse on the peak stemflow-usage day. Across all trees, an average of $11 \pm 3\%$ (± 1 standard error) of the added stemflow was used in the single day with the strongest label signature following the pulse (i.e., as calculated from the mixing model results reported in the previous subsection). This fraction of stemflow used was significantly greater than zero ($P = 0.001$). By grouping topographic position, year and species, that fraction of stemflow used was significantly

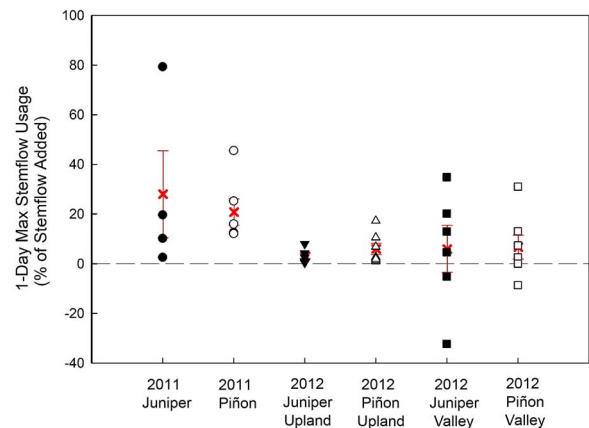


Figure 7. Individual trees (shown as circles, triangles, and squares) and mean $\pm$ 1 SE (shown as 'x's and error bars) daily usage of the added stemflow label, for piñon and juniper in 2011 and 2012 (grouped by valley and upland hillslope trees), calculated using the maximum fraction of label in trees (Figures 4 and 5) multiplied by sap flow over the same days (Figure 6).

greater than zero for the piñon in 2011, and for both the upland piñon and juniper in 2012 ($P_{\text{crit}} < 0.05$); differences between piñon and juniper were not detected in the whole dataset or its subsets.

Water potential

Predawn water potentials (Ψ_{pd}) in 2011 (Figure 8a) were less negative overall than in 2012 (Figure 8c). In 2011, Ψ_{pd} for piñon was -0.73 MPa on Day 0, and became significantly less negative on Day 2 to -0.62 MPa ($P = 0.0006$), indicating a decrease in plant water stress. Juniper Ψ_{pd} was lower than piñon at -0.86 MPa on Day 0 and showed no change that would indicate an alleviation of plant water stress throughout the experiment, as it decreased to -1.0 MPa on Day 3.

Midday water potentials (Ψ_{md}) in 2011 were more negative than Ψ_{pd} at -1.77 and -2.0 MPa for piñon and juniper, respectively. Piñon showed a change on Day 2 to -1.61 MPa ($P = 0.016$), which decreased only slightly through Day 3. Again, juniper showed no changes to indicate alleviation of water stress, instead decreasing to -2.3 MPa on Day 3 (Figure 8b).

In 2012, piñon and juniper in the valley had very similar Ψ_{pd} at about -1.0 MPa on Day 0. Upland piñon Ψ_{pd} was more negative than valley trees at -1.4 MPa, and junipers were -1.9 MPa on Day 0, indicating that the upland trees were more water stressed than the valley trees. No significant

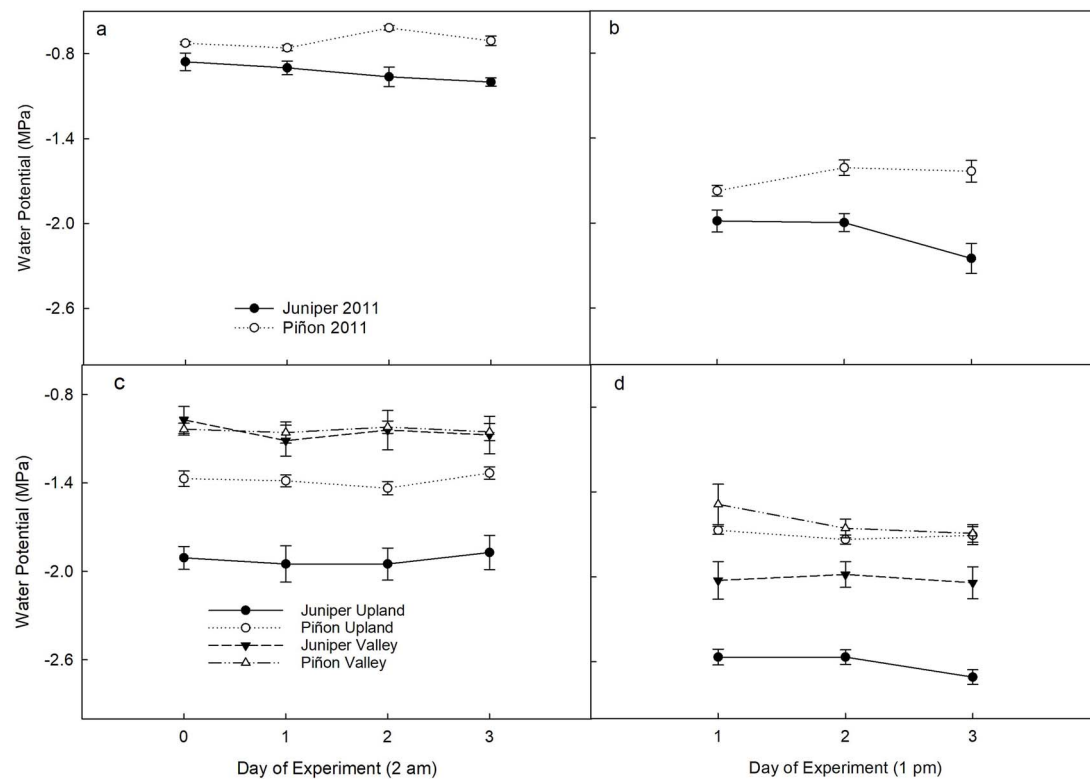


Figure 8. Mean $\pm$ 1 SE predawn (a) and midday (b) water potential by site and species in 2011. Mean $\pm$ 1 SE predawn (c) and midday (d) water potential by site and species in 2012. Stemflow was added at $\sim$ 9 a.m. on Day 1.

changes in Ψ_{pd} were seen during the experiment to suggest that stemflow alleviated water stress (Figure 8c and d).

Midday water potentials were highest for piñon in 2012, with values being similar at both upland and valley sites and ranging from -1.4 to -1.5 MPa, respectively on Day 0. The junipers had more negative values of Ψ_{md} , with values being more negative at the upland site. Juniper Ψ_{md} was about -2.0 MPa at the valley site and -2.6 MPa at the upland site on Day 0. No changes were seen in any of the trees throughout the experiment to indicate alleviation of water stress (Figure 8d).

Discussion

This is the first study to test how pulses of stemflow are taken up by plants and how that uptake affects water status and water use of the plant that generated the stemflow. It is evident that piñon and juniper have the ability to use stemflow water that would be generated by larger rainstorms in the Great Basin. Though relatively little rainfall is typically received in July for this area, this month was chosen to reduce the likelihood of multiple water sources for the trees and to maximize the potential to measure enrichment of xylem water due to application of labeled stemflow water. While the experimental conditions (i.e., stemflow generated in absence of a storm providing throughfall inputs elsewhere) are not realistic in natural settings, they allowed for specifically focusing on the question of stemflow uptake. Though the fraction of xylem water composed by stemflow was relatively small, averaging $\sim 1\%$, it was detectable. The use of stemflow water did not increase sap flow, and in only one case did it significantly increase plant water potentials.

Overall, the response in δD of juniper xylem water at all sites and in both years was more delayed than the response seen in piñon xylem water. While sap flow did not change significantly in all cases, the isotope mixing model demonstrates that all tree groups significantly took up the labeled stemflow, except for juniper in the valley in 2012 (Table 4). For all trees except the piñon in 2011 (Figure 8a), stemflow irrigation did not relieve plant water stress as measured by plant water potentials.

It is surprising that the juniper and piñon showed roughly similar fractions of stemflow in their stems because our findings suggest that the juniper may be more shallowly rooted than piñons here (Figure 3a), although others have found juniper to be more deeply rooted (West *et al.* 2007a). By using the soil water δD values and the Day 0 δD xylem water values of trees, it was shown that the piñons were using deeper water than junipers in 2011, and that the valley medium piñons were accessing deeper soil water, or possibly groundwater, in 2012 (Figure 3 and Table 3). Despite these differences, trees with deeper root uptake were similarly responsive to stemflow, which may support the double-funneling hypothesis: i.e., stemflow seems to be efficiently routed to roots despite belowground morphological differences (Johnson and Lehmann 2006, Spencer and van Meerveld 2016).

The interaction between stemflow use, bulk soil water content and root activity lead to variations in results, as wet years presumably have higher soil moisture content, more active shallow roots and more natural stemflow due to an increase in storm frequency or intensity. For example, comparing water potentials, it is clear that the trees were less stressed in the wet year. Interestingly, the wetter 2011 was the year when water potentials improved following stemflow irrigation, indicating

the importance of bulk shallow soil water for these trees. This is seen in the valley piñon, where the soils had the greatest soil water content. For some of the more stressed trees in 2012, the amount of stemflow irrigation applied (3 L on average) was greater than the amount of water used in a day, but not enough to alleviate water stress (Figure 8). A possible explanation is that the drier surface soils in 2012 resulted in stronger matric potentials and storage deficits such that they negate preferential flow routes to roots in deeper soils.

A question prompted by these results is, ‘what happened with the unused stemflow?’ If we consider that the duration of the label persisting in xylem appeared to be 1–3 days (see Figures 4 and 5), the total stemflow used may be double what we estimate for the single-day-maximum values, suggesting that ~20% of stemflow might become transpired on average; readers should note that differences between years do not only reflect differences in antecedent soil moisture conditions, but also that several trees received several times more stemflow in 2011 than they did in 2012 (see the Materials and methods section). Readers should also note that the seemingly highest mean values shown in Figure 7, i.e., those of the junipers in 2011, also had the largest standard errors and that the observed mean was not significantly different from zero ($P = 0.05$). While our findings suggest that stemflow is a water source available to trees, and that it seems to be a heavily used source to some individual trees, it remains unclear what happens with the remaining 80% of the added stemflow. One possibility is that these calculated fractions of used stemflow water are underestimates because sap flow techniques (e.g., using the empirical Granier formula, Eq. (2)) and their upscaling can be considerably uncertain (Clearwater et al. 1999, Steppe et al. 2010, Shinohara et al. 2013). However, given that we upscaled using the assumption that the entire sap flow area is equally transporting water, rather than decreasing with radial depth into the tree, we suspect that the sap flow estimates used are more likely to be overestimates than underestimates; this means that adopting a lower sap flow rate in the end-member splitting calculations (as would occur by accounting for radial sap flux variations) would decrease our estimates of the volume of the added stemflow that supplied transpiration. Another possibility is that the stemflow is used over longer time periods (exceeding the 4-day period over which we monitored), but the isotope values of the trees appear to return to the pre-stemflow values within that 4-day period (Figures 4 and 5). Stemflow may also contribute to evaporation; while some evaporation surely occurs from the stem itself, subsequent evaporation from soils likely depends on how that stemflow infiltrates when it reaches the stem-soil interface. If the influx rates exceeded infiltration capacities and generated lateral runoff, substantial evaporation from wetted surface soils and litter would be likely; no visible evidence indicated that this was the case. Alternatively, if stemflow infiltrated rapidly and percolated deeply, evaporation is unlikely to be substantial and that stemflow may preferentially reach roots but also preferentially reach deeper depths, composing deep-percolation fluxes that contribute to groundwater recharge. This deep-percolation response as well as the funneling of water to roots are both consistent with the double-funneling hypothesis (Johnson and Lehmann 2006, Spencer and van Meerveld 2016), the implications of which include that stemflow is a particularly important precipitation influx because of how it travels through the subsurface.

To estimate the percolation depth of stemflow, we did a quick calculation assuming a medium size (20 cm diameter) tree with an infiltration area to 10 cm from the bole. For a 3 L irrigation this would translate to ~3.2 cm of rain percolating to 32 cm depth below the soil surface, assuming soil porosity was 50% and field capacity was 20% of saturated capacity and ignoring preferential flow paths which are known to exist along roots of woody species (Li et al. 2009). Therefore, stemflow irrigation reached the root zone, but it is possible that fine roots are less abundant in shallower zones during the summer dry period, even if some of the labeled water was taken up by the trees (Peek et al. 2006). However, if preferential flow along roots moved some of the water deeper into the soil profile, this water may contribute to a deeper pool of soil moisture that the trees could access later as surface soil moisture becomes unavailable, especially if the preferential flow followed deep taproots.

When interpreting these results, we considered whether evaporative enrichment could have any important impact on the findings observed. It seems unlikely that evaporative enrichment would explain the observed enrichment of tree xylem water δD over the course of the experiment because evaporative enrichment would likely persist until there was another rain event. Alternatively, in this study, the tree xylem water δD values generally showed a transient increase and then a return to the pre-stemflow application Day 0 δD values. That transient response also provides confidence that biases introduced by the extraction process are not the cause of the observed stemflow signatures in xylem, because there is no reason to think that they would yield this observed transient uptake response (Chen et al. 2020). Therefore, it is reasonable to conclude that uptake of labeled stemflow produced the observed changes in δD xylem water values.

Responsiveness to stemflow inputs may reflect seasonal changes in root morphology, as investment in deep roots where water is more available could attenuate dependence on summer-time precipitation inputs (including stemflow). In nearby regions with similar climate, Peek et al. (2006) observed a reduction in the root length of Utah juniper in the upper 30 cm of soils, along with an increase in root length in deeper (>100 cm) soils from spring to summer. Research by Williams and Ehleringer (2000) suggests that root activity of two-needle piñon may be sensitive to soil temperatures, where surface roots either die, cavitate (Linton et al. 1998), or remain inactive even after summer precipitation input. A reduction in fine roots near the surface accompanied by an increase in deeper roots that tap into wetter substrates could explain why physiological responses were small (even if there were enough shallower roots that nearly all species and size classes showed a significant proportion of labeled water in xylem water). Alternatively, it may also be that the added stemflow reached those deeper water storages, but they were sufficiently large pools that the stemflow only represented a minor input; this could explain why the fraction of stemflow used for transpiration was mostly not significantly different from zero. However, we suspect that the stemflow did not infiltrate deeply and largely recharged shallow soils that still contained enough roots for the label to show up in xylem water. We hypothesize that stemflow would be more likely to have physiological effects if it occurred in a natural setting, co-occurring with precipitation that wets the bulk soil and thereby may also initiate more shallow root activity. It may

be that a bulk wetting of the soils is required for the small stemflow inputs to become funicular soil water and percolate deeply. Thus, our findings prompt us to revise our hypothesis to be that the much larger throughfall influxes are also key to the smaller stemflow influxes having notable impact; until said study is pursued, evidence for strong beneficial impacts of stemflow to plants remains scarce.

Conclusions

We hypothesized that arid-region piñon and juniper trees would use stemflow water and it would provide physiological benefits to the plants generating the stemflow. After isotopically labeled water was applied to stems to simulate stemflow, a small but significant fraction of stemflow was detectable in five out of six labeling experiments. However, in most cases, this uptake of stemflow water was not associated with a detectable change in sap flow rates or plant water potentials. Consequently, we were also unable to infer whether the physiological effects of stemflow additions differed between species or between upland and valley sites. This lack of physiological benefit may reflect that the small stemflow inputs do not significantly increase the supply of water reaching the depths where uptake is occurring. Thus, further studies could build upon our findings by pursuing long-term experiments that (i) span a range of conditions with varying soil wetness and natural precipitation events and (ii) focus on whether stemflow percolates more deeply than throughfall and thus more efficiently recharges deeper soil moisture storages that plants may rely on as shallow soils dry.

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Authors' contributions

K.A.S. conceived and designed the experiment. A.C.M. performed the experiments. K.A.S., A.C.M., and S.T.A. analyzed the data. K.A.S., A.C.M., and S.T.A. wrote the manuscript. T.K.S. provided editorial advice and project supervision.

Supplementary data

Supplementary data for this article are available at *Tree Physiology* Online.

Conflict of interest

None declared.

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Data availability

Data transparency—data available upon request to authors.

Code availability

Software application or custom code—R code for sap flow is available upon request to authors.

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